

Rahul Reddy Puchakayala

AI Product Manager

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SUMMARY

AI Product Manager with 6+ years of experience building AI, platform, and data products across fintech and healthcare. Proven track record leading 0-to-1 product development, enterprise AI platform strategy, and multi-agent systems that automate complex workflows, strengthen governance, and scale decision-making in regulated environments. Experienced in LLMs, RAG, AI evaluation, retrieval systems, and cross-functional leadership across engineering, design, risk, and compliance teams. Currently leading enterprise AI platforms at Capital One supporting 800+ users across 80+ use cases.

PROFESSIONAL EXPERIENCE

Capital One | *Product Manager, AI/ML Products*

January 2025 – Present

- Owned product strategy and roadmap for enterprise AI platforms supporting 800+ internal developer users across 80+ use cases, accelerating development-to-production cycles and achieving 89% deployment coverage across major Lines of Business (LOBs)
- Led the launch of an enterprise AI Prototyping Environment and zero-touch, self-service VectorDB platform, enabling developers to instantly deploy RAG applications while eliminating manual platform provisioning friction for engineering teams
- Defined end-to-end delivery of an LLM Fine-Tuning Service from PRD through General Availability, establishing cross-organizational working groups to standardize model customization, safety guardrails, and compliance evaluation
- Architected RAG pipelines using LLaMA and GPT-4 that cut analyst financial report generation time by 50% and reduced internal query response times by 40% across structured and unstructured data sources
- Integrated human-in-the-loop validation frameworks and NLP entity recognition models, scaling compliance monitoring from 1% to 100% of relevant transactions while improving automated fraud detection accuracy by 25%
- Defined product requirements for GPU resource optimization capabilities, improving compute utilization and accelerating access to training infrastructure for enterprise AI teams
- Led high-visibility cross-functional programs across engineering, risk, legal, and compliance to safely deploy enterprise GenAI, eliminating \$1M+ in annual third-party software spend through internally built tools

Optum (UnitedHealth Group) | *Product Manager*

August 2021 – May 2024

- Owned product strategy and roadmap for Scale Test Copilot, an AI-powered performance engineering platform serving 5,000+ internal users, reducing testing cycle times by 30%
- Identified low product adoption through customer observation and workflow analysis, redesigning anomaly prioritization with confidence scoring and severity ranking to increase adoption from near-zero to 60% within six weeks
- Led customer discovery and user research across engineering organizations, uncovering workflow bottlenecks and defining AI-driven capabilities that improved testing efficiency and operational decision-making
- Built product analytics frameworks tracking 10+ operational KPIs, enabling data-driven prioritization and improving stakeholder decision-making speed by 25%
- Partnered with engineering, UX, and data science teams to deliver AI-enabled platform capabilities, translating customer needs into PRDs, experimentation plans, and production-ready releases
- Presented roadmap strategy, customer insights, and business outcomes to senior leadership, influencing investment decisions across a platform supporting 5,000+ healthcare users

Accenture | *Data Engineer*

January 2019 – July 2021

- Designed cross-source data integration architectures using Kafka, Spark, and Hive, resolving multi-team dependencies and reducing integration times by 30%
- Analyzed large-scale streaming datasets on AWS to surface platform insights that informed business strategy across multiple client engagements, contributing to measurable top-line revenue improvements
- Translated complex operational requirements into scalable data infrastructure, improving cross-functional data accessibility and reducing decision turnaround times
- Designed and scaled ETL pipelines processing hundreds of millions of records across structured and unstructured sources, ensuring SLA compliance and improving enterprise reporting reliability

AI PRODUCT PLATFORMS

RegulatedAI — Financial & Legal Intelligence Platform | 2026

- Defined the product architecture for a multi-agent financial intelligence platform orchestrating Market Data, Risk, and Compliance workflows, reducing advisory turnaround from hours to under 15 seconds
- Designed compliance-by-architecture controls that enforce FINRA Rule 2111 and Reg BI requirements before recommendation delivery, with automated audit trails for high-risk decisions
- Built a legal research layer combining authority-weighted retrieval and a citation validation gate to reduce hallucinated citations, with an evaluation framework tracking precision and human override rates

ProductOS — AI Operating System for Product Teams | 2026

- Led the 0-to-1 development of an AI product intelligence platform that converts customer interviews, support tickets, and NPS feedback into prioritized roadmaps, PRDs, and experimentation plans in under 30 seconds, leveraging a multi-agent decision framework that reduces weeks of product discovery and planning into minutes
- Defined a schema-first ProductBrief framework that enforces evidence-backed customer pain points, source attribution, opportunity sizing, and revenue-at-risk analysis to improve decision quality and stakeholder trust

Reknew AI — Search Infrastructure for Agentic Commerce | 2025

- Defined the search and retrieval strategy for an agentic commerce platform spanning 2,250 products, benchmarking 18 embedding and retrieval configurations using NDCG@10 and LLM-as-a-Judge evaluation frameworks to improve retrieval quality by 37% over keyword-search baselines

CodeSentinel — AI-Powered Code Review Assistant | 2025

- Built and launched an AI-assisted code review platform across 10 programming languages from concept to production in under one week, reducing PR review turnaround from hours to under 10 seconds through automated security, architecture, and code quality analysis

ExecutiveAI — AI Chief of Staff | 2025

- Built an AI-powered decision intelligence platform that converts meeting transcripts into structured decision registries with automated executive review workflows, reducing leadership operational overhead by 4–6 hours per week

SKILLS

Product Management: Product Strategy, Roadmap Ownership, 0-to-1 Product Development, PRDs & User Stories, Backlog Prioritization, Agile (Scrum/Kanban), User Research, Product Analytics, KPI Definition, Experimentation & A/B Testing, Go-to-Market Strategy, Stakeholder Management

Applied AI & ML: LLMs (GPT-4, Claude, LLaMA), RAG Systems, Multi-Agent Systems (LangGraph), AI Evaluation Frameworks, LLM-as-a-Judge, AI Governance, Human-in-the-Loop Workflows, Prompt Engineering, NLP, XGBoost, Vector Search (ChromaDB)

Platforms & Infrastructure: AWS, GCP, Azure, Vertex AI, Kubeflow, Kafka, Spark, REST APIs, CI/CD, GitHub Actions

Programming & Developer Tools: Python, SQL, Claude Code, Cursor, GitHub Copilot, Lovable

Analytics & Product Tools: Power BI, Tableau, Jira, Confluence, Figma, Aha!

EDUCATION

University of Texas at Dallas, U.S.A May 2026

Master of Science, Information Technology & Management

- **Honors & Awards:** Dean's Excellence Scholarship recipient ; Recipient of the OWLIE Award for Graduate Organization of the Year 2025

LEADERSHIP & COMMUNITY

The Product Base Club, UTD | President Dec 2024 – May 2026

- Led a cross-functional team of 8 to grow UTD's largest product management community to 500+ members, increasing active participation by 40% through 15+ industry-led workshops on AI, product strategy, and user research
- Recognized with the OWLIE Award as JSOM Graduate Organization of the Year (2025) among 50+ student organizations for driving exceptional member engagement, professional development, and sustained community growth